Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

1 Define the following:

Let Ω be a non-empty set, $\mathcal{F}$ is a collection of subsets of Ω and $P : \mathcal{F} \to \mathbb{R}$ is a function.

Define $\mathcal{F}$. P : $\mathcal{F} \to \sigma$-algebra ; $P \to$ probability measure on $(\Omega, \mathcal{F})$

- (1) $(\Omega, \mathcal{F}, P)$ is called a probability space if ...

- (2) A function X, defined on a probability space $(\Omega, \mathcal{F}, P)$ is called a random variable (r.v) if ...

Let X be a random variable defined on $(\Omega, \mathcal{F}, P)$, with c.d.f F_X and law $P_X \equiv \mu_{F_X}$.

- (3) X is a *Exponential*(λ) random variable ($\lambda > 0$ is a constant) if ...

- (4) X is called a *singular* variable if ...

Let $\{X_n\}, X$ be random variables defined on some $(\Omega, \mathcal{F}, P)$.

- (5) $\{X_n\}$ is a sequence of independent and identically distributed (i.i.d.) random variables if ... $\forall m \neq n$ $P_{X_m} = P_{X_n}$

✓ - (6) X_n converges to X "almost surely (a.s)" or "with probability 1 (w.p. 1)" or "a.e (P)" if ...

- (7) X_n converges to X in probability (w.r.t P) if ...

- (8) if each $X_n \in L^1(\Omega, \mathcal{F}, P)$, then the collection $\{X_n\}$ is called uniformly integrable (UI) if ...

(3×8 = **24** points)

2 Let X, Y be two **independent** random variables on some probability space $(\Omega, \mathcal{F}, P)$, with marginal laws P_X, P_Y on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Recall that X and Y independent iff their joint law is the *product measure* of their marginal laws, i.e $P_{(X,Y)} = (P_X) \times (P_Y)$ on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$. Let m is the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and $m^2 \doteq m \times m$ is the Lebesgue measure on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$. Let f, g be two non-negative functions in $L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$.

(a) (b)
Tonelli'
Fubini
f(x,y) can be factorized into g(x) h(y)+
f : function on (R*R to R)

- (a) If X and Y are independent and $E(X^2) < \infty$, $E(Y^2) < \infty$, then carefully prove

prove $EXY = EXEY$.

$$Var(X+Y) = Var(X) + Var(Y).$$

$P_Y(A_2) = \int_{A_2} g \, dm$

non-negative measurable

- (b) Suppose $P_X \ll m, P_Y \ll m$, with $\frac{dP_X}{dm} = f$ and $\frac{dP_Y}{dm} = g$. Then prove

$$P_{(X,Y)} \ll m^2, \quad \frac{dP_{(X,Y)}}{dm^2} = f.g \quad a.e\ (m^2)$$

$P_{(X,Y)}(A) = \int_A h \, dm^2$

$= \int_{A_1} (\int_{A_2} h \, dm) \, dm$

$\forall x_n \to x$, $f(x_n) \to f(x)$.

- (c) If f is continuous and bounded on $\mathbb{R}$, so is $f * g$.

$h(x,y)$. $x \in \mathbb{R}$, $y \in \mathbb{R}$

- (d) If $\lambda \doteq P_X * P_Y$ Prove $\frac{d\lambda}{dm} = f * g$ a.e (m), f and g are as in (b).

- (e) If Z is a random variable with $P_Z = \lambda$ as in (d) above, then how is Z related to X and Y ? What does the result in (d) say about the law of Z ?

(6+6+6+6+(3+3)= **30** points)

$P_{X,Y}(A) = (P_X \times P_Y)(A) \to A_1 \times A_2$

$= \int_{A_1 \times A_2} dP_X \, dP_Y = \int_{A_1} (\int_{A_2} dP_Y) \, dP_X$

$= \int_{A_1} g \, dm \, dP_X = fg$

$\limsup_{n\to\infty} \int \lim_{k\to\infty} X_{nk}\, dP$

$\lim_{k\to\infty} \int X_{nk}\, dP$

$\liminf I_{A_n}$

$\limsup \int I_{A_n}\, dP$

$P(A_n)$

$X_{nk} \uparrow X_n$

3 Let $\{X_n\}_{n\geq 1}$ be a sequence of *nonnegative* random variables defined on a probability space $(\Omega, \mathcal{F}, P)$. Let

$$Y(\omega) = \liminf_{n\to\infty} X_n(\omega),\ \omega \in \Omega.$$

$\liminf_{n\to\infty} (\lim_{k\to\infty} X_{nk}) = \lim_{n\to\infty} (\liminf_{k\to\infty} X_{nk})$

(i) Prove $Y : \Omega \to I\!R$ is a random variable.

(ii) If $\limsup_{n\to\infty} E(X_n) < \infty$, then prove that $Y \in L^1(\Omega, \mathcal{F}, P)$.

(iii) If $M(t)$ is the moment generating function of Y, i.e. $M(t) = E(e^{tY})$, $t \in I\!R$, then show

$$P(Y > a) \leq \inf_{t>0} e^{-ta} M(t), \quad a > 0.$$

(5+5+5= **15** points)

4 Let m be the Lebesgue measure on $[0,1]$ and $I_A(x)$ denote the indicator function of A. Consider the following sequence of random variables on $(\Omega = [0,1], \mathcal{F} = \mathcal{B}[0,1], P = m)$: For $n \geq 1$, $X_n(\omega) = nI_{(0,\frac{1}{n^2})}(\omega)$ and $X(\omega) = 0$ for $\omega \in \Omega$.

i : for w is fixed

(i) Show $X_n \to X$ almost surely (P).

(ii) (iii)TX:先不取Limit, n is treated as fixed

(ii) Does $X_n \to X$ in probability ?

(iii) Show $X_n \to X$ in $L^p \equiv L^p(\Omega, \mathcal{F}, P)$ for $0 < p < 2$, but this convergence does not hold for $2 \leq p < \infty$.

(iv) Show that $X_n \to X$ in L^p for some $0 < p < \infty$ implies that $X_n \to X$ in L^q, for any $0 < q < p$.

(3+3+(3+2)+5 = **16** points)

5 Let $\alpha \in I\!R, \beta > 0$ be constants.

(i) Show that a $Normal(\alpha, \beta^2)$ random variable exists: $C = \{(-\infty, x) : x \in I\!R\}$ semi-algebra. μ. μ_F

In other words, using steps of the Caratheodory extension theorem, exhibit a probability space $(\Omega, \mathcal{F}, P)$ and a random variable $X : \Omega \to I\!R$, such that its c.d.f. is given by Φ where

$F = \int_C (\ )\, d\mu$ $\quad P_X = \Phi(x)$ $\quad \mu_F = P_X$ $\quad P \circ X^{-1}$ $\quad \mu \quad \mu_F^*$ $\quad \mu_F\ \sigma(C)$ $\quad \mu_{\mu_F^*} \supset \sigma(C)$ $\quad \mathcal{B}(I\!R)$ $\quad \mu_F^* = \mu_F$ on C

$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\beta^2}} e^{-\frac{(u-\alpha)^2}{2\beta^2}}\, du,\ x \in I\!R.$$

(ii) Compute the following Lebesgue integral (Clearly stating any result used)

$$\int_\Omega X(\omega)\, dP(\omega).$$

(iii) For each $n \geq 1$, let μ^n be a probability measure on $(I\!R^n, \mathcal{B}(I\!R^n))$. Under what conditions can we say that there is a probability measure P on $(I\!R^\infty, \mathcal{B}(I\!R^\infty))$ such that $P(A \times I\!R^\infty) = \mu^n(A)$ for any $A \in \mathcal{B}(I\!R^n)$, for any $n \geq 1$? Describe a sequence of random variables $\{X_n : n \geq 1\}$ defined on $(\Omega = I\!R^\infty, \mathcal{F} = \mathcal{B}(I\!R^\infty), P)$ such that the joint distribution of $(X_1, \ldots, X_n)$ is given by μ^n, for all $n \geq 1$.

(iv) Using part (iii) or otherwise, show that there exists $\{X_n : n \geq 1\}$ which is an i.i.d. sequence of $Normal(\alpha, \beta^2)$ random variables.

(5+5+5+5= **20** points)

TABLE 4.6.1. Some discrete univariate distributions.

Mean μ	$\mu(A),\ A \in \mathcal{B}(\mathbb{R})$
Bernoulli (p), $0 < p < 1$	$\sum_{i=0}^{1} p^i (1-p)^{1-i} I_A(i)$
Binomial (n, p), $0 < p < 1$, $n \in \mathbb{N}$	$\sum_{i=0}^{n} \binom{n}{i} p^i (1-p)^{n-i} I_A(i)$
Geometric (p), $0 < p < 1$	$\sum_{i=0}^{\infty} p(1-p)^i I_A(i)$
Poisson (λ), $0 < \lambda < \infty$	$\sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} \cdot I_A(i)$

TABLE 4.6.2. Some standard absolutely continuous distributions. Here m denotes the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Measure μ	$\mu(A),\ A \in \mathcal{B}(\mathbb{R})$
Uniform (a, b), $-\infty < a < b < \infty$	$m(A \cap [a, b])/(b - a)$
Exponential (β), $\beta \in (0, \infty)$	$\int_A \frac{1}{\beta} \exp(-x/\beta) I_{(0,\infty)}(x) m(dx)$
Gamma (α, β), $\alpha, \beta \in (0, \infty)$	$\int_A \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp(-x/\beta) I_{(0,\infty)}(x) m(dx)$ where $\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx$, $a \in (0, \infty)$
Beta (α, β), $\alpha, \beta \in (0, \infty)$	$\frac{1}{B(\alpha,\beta)} \int_A x^{\alpha-1}(1-x)^{\beta-1} I_{(0,1)}(x) m(dx)$, where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a + b)$, $a, b \in (0, \infty)$
Cauchy (γ, σ) $\gamma \in \mathbb{R}$, $\sigma \in (0, \infty)$	$\int_A \frac{1}{\pi\sigma} \frac{\sigma^2}{\sigma^2 + (x-\gamma)^2} m(dx)$
Normal (γ, σ^2), $\gamma \in \mathbb{R}$, $\sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \exp(-(x-\gamma)^2 / 2\sigma^2) m(dx)$
Lognormal (γ, σ^2), $\gamma \in \mathbb{R}$, $\sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \frac{e^{-(\log x - \gamma)^2/2\sigma^2}}{x} I_{(0,\infty)}(x) m(dx)$

1. Solution:

   (1) $\Omega \in \mathcal{F}; A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}; \{A_n\} \in \mathcal{F} \Rightarrow \cup_{n\geq 1} A_n \in \mathcal{F}$ and $P(\emptyset) = 0, P(\Omega) = 1, \{A_n\}_{n\geq 1}$ are disjoint $\Rightarrow P(\cup_{n\geq 1} A_n) = \sum_{n\geq 1} P(A_n)$.

   (2) $X : \Omega \to \mathbb{R}$ and $\forall\ B \in \mathcal{B}(\mathbb{R}),\ X^{-1}(B) \in \mathcal{F}$, i.e. X is a real-valued measurable transformation.

   (3) X is a (real valued) random variable with law P_X given by *Exponential*(λ) probability on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, i.e. $P_X(A) = \int_A \lambda e^{-\lambda u} dm(u)$.

   (4) $F'_X(x) = 0$ a.e. (m) or $P_X \perp m$.

   (5) each X_n is a random variable with the same distribution, i.e. $P_{X_n} = P_{X_m}, \forall\ n, m$, and $\{X_i : i \geq n\}$ is an independent sequence, i.e. $P(X_{\alpha_1} \in B_1, \cdots, X_{\alpha_k} \in B_k) = \prod_{i=1}^{k} P(X_{\alpha_i} \in B_i), \forall\ \{\alpha_1, \cdots, \alpha_k\} \subset \mathcal{B}(\mathbb{R}), i = 1, \cdots, k, k \in \mathbb{N}$.

   (6) $\exists\ A$ s.t $P(A) = 1$ and $\forall\ \omega \in A, X_n(\omega) \to X(\omega)$ as $n \to \infty$, i.e. $P(X_n \to X) = 1$.

   (7) $P(|X_n - X|) > \epsilon \to 0$ as $n \to \infty$, $\forall\ \epsilon > 0$.

   (8) $\lim_{t\to\infty} \sup_{n\geq 1} \int_{\{|X_n|>t\}} |X_n| dP = 0$, i.e. $\lim_{t\to\infty} \sup_{n\geq 1} E(|X_n| I[|X_n| > t]) = 0$.

2. Solution:

   (a) Note that

$$\begin{aligned} Var(X+Y) &= E([(X-EX)+(Y-EY)]^2) \\ &= Var(X) + Var(Y) + 2E[(X-EX)(Y-EY)] \\ &= Var(X) + Var(Y) + 2[E(XY) - EXEY] \end{aligned}$$

Now $E|XY| \leq \sqrt{EX^2EY^2} < \infty$ by Cauchy-Schwarz inequality. So $XY \in L^1(\Omega, \mathcal{F}, P)$. Then, by using Fubini theorem,

$$\begin{aligned} E(XY) &= \int_\Omega X(\omega)Y(\omega)dP(\omega) \\ &= \int_{\mathbb{R}^2} xy\ d(P_X \times P_Y)(x, y) \\ &= \int_{\mathbb{R}} x\ dP_X(x) \int_{\mathbb{R}} y\ dP_Y(y) \\ &= \int_\Omega X(\omega)\ dP_X(\omega) \int_\Omega Y(\omega)\ dP_Y(\omega) \\ &= EXEY \end{aligned}$$

Hence $Var(X+Y) = Var(X) + Var(Y)$.

(b) From $\frac{dP_X}{dm} = f$, $\frac{dP_Y}{dm} = g$, we have

$$\begin{aligned}
P_{(X,Y)}(A_1 \times A_2) &= (P_X \times P_Y)(A_1 \times A_2) \text{ (by definition of product measures)} \\
&= P_X(A_1)P_Y(A_2) \text{ (by definition of RN derivative)} \\
&= \left(\int I_{A_1} f dm\right)\left(\int I_{A_2} g dm\right) \text{ (by Tonelli's theorem since } f, g \geq 0) \\
&= \int I_{A_1}(x) I_{A_2}(y) f(x) g(y)\ dm(x) dm(y) \\
&= \int_{A_1 \times A_2} f(x)g(y)\ dm^2(x,y)
\end{aligned}$$

Since on rectangles $A_1 \times A_2$ on $\mathbb{R}^2$,

$$P_{(X,Y)}(A_1 \times A_2) = \int_{A_1 \times A_2} f(x)g(y)\ dm^2(x,y),$$

By uniqueness of Caratheodory extension theorem, we have

$$P_{(X,Y)}(A) = \int_A fg\ dm^2, \quad \forall A \in \mathcal{B}(\mathbb{R}^2)$$

Note: another argument is the following

$$\begin{aligned}
(P_X \times P_Y)(A) &= \int P_Y(A_{1x})\ dP_X(x) \\
&= \int \left[\int I_{A_{1x}}\ dP_Y(y)\right]\ dP_X(x) \\
&= \int \left[\int I_{A_{1x}} g(y)\ dm(y)\right]\ dP_X(x) \\
&= \int \int I_{A_{1x}}(y) g(y)\ dm(y) f(x) dm(x) \\
&= \int \int I_A(x,y) f(x) g(y)\ dm^2(x,y)
\end{aligned}$$

[More rigorous proof is: define $\mu(A) = \int_A fg\ dm^2$, $\forall A \in \mathcal{B}(\mathbb{R}^2)$, then $\mu = P_{(X,Y)}$ on $\mathcal{C} = \{A \times B : A, B \in \mathcal{B}(\mathbb{R}^2)\}$, and it is a finite measure on $\mathcal{C}$. Hence by uniqueness theorem, $\mu = P_{(X,Y)}$ on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$.]

From this, it is clear that $P_{(X,Y)} \ll m^2$ and $\frac{dP_{(X,Y)}}{dm^2} = f \cdot g$ a.e.(m^2).

(c) Note that bounded. $\sup_{x \in \mathbb{R}} |f(x)| = M < \infty$ and $(f * g)(x) = \int f(x-u)g(u)\ du$.

If $x_n \to x$, since f is continuous and $\phi(x) = x - u$ is continuous for fixed u, then $K_n(u) = f(x_n - u) \to f(x-u)$. So for all $u \in \mathbb{R}$,

$$K_n^x(u) \triangleq f(x_n - u)g(u) \to K^x(u) \triangleq f(x-u)g(u)$$

Then $|K_n^x(u)| \leq Mg(u)$ and $g \in L_1$. By DCT, we have

$$\int K_n^x(u)\,du \to \int K^x(u)\,du$$

as $n \to \infty$, i.e. $(f*g)(x_n) = \int f(x_n-u)g(u)\,du \to \int f(x-u)g(u)\,du = (f*g)(x)$. So $\forall\, x_n \to x$, $(f*g)(x_n) \to (f*g)(x)$, i.e. $f*g$ is continuous.
Also note that for all x,

$$\int |f(x-u)g(u)|\,du \leq M\int |g(u)|\,du \equiv M' \leq \infty.$$

So for all x,

$$|(f*g)(x)| \leq \int |f(x-u)g(u)|du \leq M'$$

Hence $f*g$ is bounded.

(d) If $P_X \ll m, P_Y \ll m$, then we have $f = \frac{dP_X}{dm}, g = \frac{dP_Y}{dm}$ by the definition of Radon-Nikodym derivatives, the following holds for $A \in \mathcal{B}(I\!R)$ (proving the claim):

$$\begin{aligned}
(P_X * P_Y)(A) &= \int\int I_A(x+y)\,dP_X(x)dP_Y(y) \\
&= \int\left(\int I_A(x+y)f(x)dm(x)\right)dP_Y(y) \quad \text{(by change of variable)} \\
&= \int\left(\int I_A(u)f(u-y)dm(u)\right)dP_Y(y) \quad \text{(by Fubini's theorem)} \\
&= \int I_A(u)\left(\int f(u-y)dP_Y(y)\right)dm(u) \\
&= \int I_A(u)\left(\int f(u-y)g(y)dm(y)\right)dm(u) \\
&= \int I_A(u)(f*g)(u)dm(u) \\
&= \int_A (f*g)(u)dm(u)
\end{aligned}$$

Hence by the uniqueness of the RN derivatives, $\frac{\lambda}{dm} = f*g$ a.e. (m).

(e) It says Z has the same law as $X+Y$. Part (d) says if X, Y has densities f, g respectively, then Z will have density $f*g$.

3. Solution:

(i) Note $Y(\omega) = \lim_{n\to\infty}[\inf_{m\geq n} X_m(\omega)] = \sup_{n\geq 1}[\inf_{m\geq n} X_m(\omega)]$, $\omega \in \Omega$. Define $Y_n(\omega) = \inf_{m\geq n} f_m(\omega)$, $\omega \in \Omega, n \geq 1$. For each $a \in \mathbb{R}$

$$\{\omega : Y_n(\omega) \geq a\} = \cap_{m\geq n}\{\omega : X_m(\omega) \geq a\} = \cap_{m\geq n} X_m^{-1}([a,\infty)) \in \mathcal{F}$$

since all X_n's are $\mathcal{F}$-measurable. This proves Y_n is $\mathcal{F}$-measurable for each $n \geq 1$. Hence, for each $b \in \mathbb{R}$

$$\{\omega : Y(\omega) \leq b\} = \{\omega : \sup_{n\geq 1} Y_n(\omega) \leq b\} = \cap_{n\geq 1}\{\omega : Y_n(\omega) \leq b\} = \cap_{n\geq 1} Y_n^{-1}((-\omega, b]) \in \mathcal{F}.$$

This proves that Y is $\mathcal{F}$-measurable.

(ii) Note that by $\boxed{\text{Fatou's lemma,}}$

$$\begin{aligned}\int Y(\omega)\, dP = \int \liminf_{n\to\infty} X_n(\omega)\, dP &\leq \liminf_{n\to\infty} \int X_n(\omega)\, dP \\ &\leq \limsup_{n\to\infty} \int X_n(\omega)\, dP \\ &< \infty\end{aligned}$$

So $Y \in L^1(\Omega, \mathcal{F}, P)$.

(iii) Note that $P(X > a) = P(e^{tX} > e^{ta})$ since $f(x) = e^{tx}$ is increasing function for $t > 0$.
Markov inequality
Then $E[e^{tX}] = E[e^{tX} I(X \leq a) + e^{tX} I(X > a)] \geq e^{ta} P(X > a)$. Thus $P(X > a) \leq e^{-ta} M(t)$ for all $t > 0$. So $P(X > a) \leq \inf_{t>0} e^{-ta} M(t),\ a > 0$.

4. Solution:

(i) $\forall\ \omega \in (0,1), X_n(\omega) = 0$ for large n since $\frac{1}{n^2} < \omega$ for $n \geq \frac{1}{\sqrt{\omega}}$ if $\omega > 0$. Also we have $X_n(0) = X_n(1) = 0, \forall\ n \geq 1$. So $X_n(\omega) \to X(\omega), \forall\ \omega \in [0,1]$, i.e. $X_n \to X$ a.e. (P).
and $P([0,1]) = 1$

(ii) $m(\{\omega : |X_n(\omega) - X(\omega)| > \epsilon\}) = m((0, \frac{1}{n^2})) = \frac{1}{n^2} \to 0$ as $n \to \infty$, $\forall\ \epsilon > 0$. So $X_n \xrightarrow{P} X$.

(iii) $(E|X_n - X|^p)^{\frac{1}{p}\wedge 1} = (\int |X_n - X|^p\, dP)^{\frac{1}{p}\wedge 1} = (n^p \cdot m((0, \frac{1}{n^2})))^{\frac{1}{p}\wedge 1} = n^{\frac{p-2}{p\vee 1}} \to 0$ as $n \to \infty$ iff $0 < p < 2$. Hence $X_n \to X$ in L^p for $0 < p < 2$, but this convergence doesn't hold for $2 \leq p < \infty$.

(iv) Let $g(x) = x^{\frac{p}{q}}, x > 0$ and $\frac{p}{q} > 1$, then $g(x)$ is a convex function. Since $X_n \to X$ in L^p, then $(\int |X_n - X|^p dP) \to 0$ as $n \to \infty$. By Jensen's inequality, we have

$$g\left(\int |X_n - X|^q dP\right) \leq \int g(|X_n - X|^q) dP$$

$$\left(\int |X_n - X|^q dP\right)^{\frac{p}{q}} \leq \int |X_n - X|^p dP$$

Then

$$\int |X_n - X|^q dP \leq \left(\int |X_n - X|^p dP\right)^{\frac{q}{p}} \to 0, \quad \text{as } n \to \infty.$$

i.e. $(\int |X_n - X|^q dP)^{\frac{1}{q}\wedge 1} \to 0$ as $n \to \infty$. Hence $X_n \to X$ in L^q for any $0 < q < p$.

5 Solution:

(i) Define $\mathcal{C} = \{(a,b], (b,\infty) : -\infty \le a, b < \infty\}$ and $\mu_\Phi((a,b]) = \Phi(b) - \Phi(a)$, $\mu_\Phi((b,\infty)) = 1 - \Phi(b)$. This defines a measure on the semi-algebra $\mathcal{C}$. Caratheodory extension theorem states that there exists an extension μ^* of μ_Φ to $(\mathbb{R}, \mathcal{M})$, and $\mathcal{C} \subset \mathcal{M}$, where

$$\mu^*(A) = \inf\left\{\sum_{n=1}^{\infty} \mu_\Phi(A_n) : \{A_n\} \subset \mathcal{C}, A \subset \cup_{n=1}^{\infty} A_n\right\}, \ A \subset \mathbb{R},$$

and $\mathcal{M}$ is the set of μ^*-measurable sets, i.e. $\mathcal{M} = \{A \subset \mathbb{R} : \mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c), \ \forall E \subset \mathbb{R}\}$. Since $\mathcal{C} \subset \mathcal{M}$, and we know that $\mathcal{B}(\mathbb{R}) = \sigma\langle\mathcal{C}\rangle$, then we have $\mathcal{B}(\mathbb{R}) \subset \mathcal{M}$. Hence, we get an extension of μ_Φ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ by restricting μ^* to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. We will denote this measure as μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Define $(\Omega, \mathcal{F}, P) = (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu)$. Define $X : \Omega \to \mathbb{R}$ as $X(\omega) = \omega$. Clearly, it is measurable and μ is a probability (and hence X is a random variable). Also, the law of X is $P_X(A) = P(X^{-1}(A)) = \mu(A)$ for all $A \in \mathcal{B}(\mathbb{R})$. $F_X(x) = P(X \le x) = \mu((-\infty, x]) = \Phi(x)$, for all $x \in \mathbb{R}$.

(ii) Here we use the following change of variable formula for Lebesgue integrals: If X is a measurable transformation from $(\Omega, \mathcal{F}, P)$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then $X \in L^1(\Omega, \mathcal{F}, P)$ iff $I \in L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), P_X)$, where $I(x) = x$, for $x \in \mathbb{R}$, and in that case,

$$\int X dP = \int_{\mathbb{R}} I(x) dP_X.$$

From the definition of P_X in the problem and relationship between Riemann and Lebesgue integrals, we know that the right side of the above equation is

$$\int_{\mathbb{R}} I(x) dP_X = \int_{\mathbb{R}} I(x) \frac{1}{\sqrt{2\pi\beta^2}} e^{-\frac{(x-\alpha)^2}{2\beta^2}} dm(x) = \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi\beta^2}} e^{-\frac{(x-\alpha)^2}{2\beta^2}} dx,$$

where is the last expression is a usual Riemann integral. We know from elementary calculus that this integral is equal to α.

(iii) From Kolmogorov's Consistency theorem, only condition we need here is that $\mu^{(n+1)}(A \times \mathbb{R}) = \mu^n(A)$ for all $A \in \mathcal{B}(\mathbb{R}^n)$ for all $n \ge 1$. For the sequence of random variables, define for $\omega = (\omega_1, \omega_2, \ldots) \in \mathbb{R}^\infty$, $X_n(\omega) = \omega_n$, $n \ge 1$. Then the joint law of $(X_1, \ldots, X_n)$ is given by

$$P_{(X_1,\ldots,X_n)}(A) = P((X_1, \ldots, X_n) \in A) = P(A \times \mathbb{R}^\infty) = \mu^n(A), \text{ for } \forall\, A \in \mathcal{B}(\mathbb{R}^n),\ n \ge 1.$$

(iv) In part **(i)**, define $\mu^{(1)} = \mu \equiv N(\alpha, \beta^2)$ and $\mu^{(n+1)} = \mu^{(n)} \times \mu$, for $n \ge 1$. That is, for each $n \ge 1$, μ^n is the product measure generated by μ on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. Then for each $n \ge 1$, μ^n is a probability on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ and $\mu^{(n+1)}(A \times \mathbb{R}) = \mu^n(A)$ for $A \in \mathcal{B}(\mathbb{R}^n)$. Hence, as in part **(i)**, we can construct a sequence of random variables $\{X_n : n \ge 1\}$ on some probability space $(\Omega, \mathcal{F}, P)$ such that joint distribution of $(X_1, \ldots, X_n)$ is given by μ^n, which is the product measure generated by μ. Hence $\{X_n : n \ge 1\}$ is an i.i.d. sequence.